1. Ultrsonic Sensor with LED

```
int trigPin = 9:
int echoPin = 8;
int led_Pin = 3;
int distance_threshold = 40;
float duration_us, distance_cm;

void setup () {
Serial.begin (9600);
pinMode(trigPin, OUTPUT);
pinMode(echoPin, INPUT);
pinMode(led_Pin, OUTPUT);
}

void loop () {
digitalWrite(trigPin, HIGH);
delayMicroseconds (10);
digitalWrite(trigPin, LOW);

duration_us = pulseIn (echoPin, HIGH);

distance_cm = 0.017 * duration_us;
if (distance_cm < distance_threshold)
digitalWrite(led_ Pin, HIGH);
else digitalWrite(led_Pin, LOW);

Serial.print("distance: ");
Serial.print (distance_cm);
Serial.println(" cm");
delay (500);
}
```

2. Temp & Humidity with LCD

```
#include "DHT.h"
#include<LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 16, 2);
#define DHTPIN 2
#define DHTTYPE DHT11
DHT dht (DHTPIN, DHTTYPE);
void setup () {
}
void loop() {
float humi=dht.readHumidity (0);
float tempC=dht.readTemperature();
Serial.print ("humidity: ");
Serial.print (humi);
Serial.print ("%");
Serial.print ("        ");
Serial.print ("temperature");
Serial.print (tempC);
Serial.print ("     .C    ");

lcd.init();
lcd.backlight();
lcd.setCursor(0,0);
lcd.print("humi: ");
lcd.print(humi);
lcd.print ("%");
Serial.begin (9600);
dht.begin();
lcd.setCursor (0,1);
lcd-print("Temp: ");
lcd-print(tempC);
lcd-print ("      .C     ");
delay (500);
}
```

3. Flame Sensor

```
int Buzzer =13;
int FlamePin = 2;
int Flame = HIGH;
void setup() {
pinMode (Buzzer, OUTPUT);
pinMode (FlamePin, INPUT);
Serial.begin(9600);
}

void loop() {

Flame = digitalRead (FlamePin);
if (Flame == HIGH)
     {
Serial.println("HIGH FLAME");
digitalWrite (Buzzer, HIGH);

}
else
{
Serial.println("NO FLAME");
digitalWrite(Buzzer, LOW);
}
}
```

4. Water Level Sensor

```
int FloatSensor-2;
int led=13;
int buttonState=1;
void setup() {
Serial.begin(9600);
pinMode (FloatSensor, INPUT_PULLUP);
pinMode (led, OUTPUT);
}

void loop() {
buttonState=digitalRead(FloatSensor);
if (buttonState==HIGH) {
digitalWrite(led, LOW);
Serial.println("WATERLEVEL HIGH");
}
else{
digitalWrite(led, HIGH);
Serial.println("WATERLEVEL -LOW");
}
delay(1000);
}
```

5. Moisture Sensor

```
int sensor_pin = A0; // Soil Sensor input at Analog PIN AO
int output_value;
void setup() {
Serial.begin(9600);
Serial.println("Reading From the Sensor ...");
delay(2000);
}
void loop() {
output_value= analogRead(sensor_pin);
output_value = map(output_value, 500,10,0,100);
Serial.print("Mositure: ");
Serial.print(output_value);
Serial.println("%");
delay(1000);
}
```

Interfacing LED

```
int led_pin =13;
int sensor_pin =8;
void setup() {
pinMode(led_pin, OUTPUT);
pinMode(sensor_pin, INPUT);
}

void loop() {
if(digitalRead(sensor_pin) == HIGH){
digitalWrite(led_pin, HIGH);
} else {
digitalWrite(led_pin, LOW);
delay(1000);
}}
```

10. IR Sensor

```
#include <Servo.h>
#include <Servo.h>
Servo tap_servo;
int sensor_pin = 4;
int tap_servo_pin =5;
int val;
void setup() {
pinMode(sensor_pin,INPUT);
tap_servo.attach(tap_servo_pin);
}
void loop() {
val = digitalRead(sensor_pin);
if (val==0) {
tap_servo.write(0);
}
if (val==1)
{       tap_servo.write(180);

}}
```

8. LDR Sensor

```
const int ledPin=2;
const int ldrPin=A0;
void setup()
{
Serial.begin(9600);
pinMode(ledPin, OUTPUT);
pinMode(ldrPIn, INPUT);
}
void loop()
{
int ldrStatus = analogRead(ldrPin);
if (ldrStatus <= 200)
{
digitalWrite(ledPin, LOW);
Serial.print("Its BRIGHT, Turn OFF the LED : ");
Serial.println(ldrStatus);
}
else
{
digitalWrite(ledPin, HIGH);
Serial.print("Its DARK, Turn ON the LED : ");
Serial.println(ldrStatus);
}
}
```

6. Touch Sensor

```
const int TOUCH_SENSOR_PIN = 2; // the Arduino's input pin that connects to the sensor's SIGNAL pin
const int LED PIN = 3;
void setup() {
Serial.begin(9600);
pinMode (TOUCH_SENSOR_PIN, INPUT); // initialize the Arduino's pin as an input
pinMode(LED_PIN, OUTPUT);
}
void loop() {
int touchState = digitalRead(TOUCH_SENSOR_PIN);
if (touchState == HIGH)
{
Serial.println(" The Sensor being touched");
digitalWrite(LED_PIN, HIGH);
delay(1000);
}
else
{
if (touchState == LOW)
{
Serial.println(" The Sensor is untouched");
digitalWrite(LED_PIN, LOW);
delay(1000);
}
}
}
```

7. Door Sensor

```
const int door_sensor_pin=13; // Arduino pin connected to door sensor's pin
const int led_pin=3;
int doorstate;
void setup() {
Serial.begin(9600);        // initialize serial
pinMode (door_sensor_pin, INPUT_PULLUP); // set arduino pin to input pull-up mode
pinMode(led_pin, OUTPUT);
}
void loop() {
doorstate-digitalRead(door_sensor_pin); // read state
if doorstate==HIGH)
{
Serial.println("The door is open");
digitalWrite(led_pin, HIGH);
delay(1000);
}
else
{
Serial.println("The door is closed");
digitalWrite(led_pin, LOW);
delay(1000);
}
}
```

9. Smoke Sensor

```
int redLed=12;
int greenLed=11;
int buzzer=10;
int smokeA0=A5;
int sensorThres=150; // Your threshold value
void setup() {
// put your setup code here, to run once:
pinMode(redLed, OUTPUT);
pinMode(greenLed, OUTPUT);
pinMode(buzzer, OUTPUT);
pinMode(smokeA0, INPUT);
Serial.begin(9600);
}
void loop() {
// put your main code here, to run repeatedly:
int analogSensor=analogRead(smokeA0);
Serial.print("Pin A0");
Serial.println(analogSensor);
if (analogSensor>sensorThres) // Checks if it has reached the thr
{
digitalWrite(redLed,HIGH);
digitalWrite(greenLed,LOW);
tone(buzzer,1000,200);
}
else
{
digitalWrite(redLed,LOW);
digitalWrite(greenLed,HIGH);
noTone(buzzer);
}
delay(100);
}
```






























